GENERATIVE AI FOR MUSIC - PRACTICAL WORK 1

Diognei de Matos
Federal University of Minas Gerais
diogneimatos@dcc.ufmg.br

ABSTRACT

This work presents a genetic algorithm for MIDI music generation. The method defines a musical structure inspired by Brazilian popular music, with eight instruments and an arrangement comprising an introduction, two sequences of main sections and choruses, and an ending. Its harmonic structure uses eight distinct chords: four for the main sections and four for the choruses. Results show that the model generates diverse songs in this style, with progressive improvements driven by the genetic algorithm.

1. INTRODUCTION

Algorithmic music generation creates musical compositions through computational procedures, enabling automated composition under explicitly defined musical structures and constraints. Classical approaches include generative grammars, Markov chains, and genetic algorithms. In this work, the choice of genetic algorithms was motivated by their recent applications to music generation and their ability to iteratively improve candidate solutions through evolution. The system operates in the symbolic domain, generating reproducible MIDI files that can be rendered as audio for qualitative evaluation.

The proposed system applies genetic evolution to generate complete musical pieces inspired by Brazilian popular music. Each composition uses eight instruments and follows a predefined structure comprising an introduction, two sequences of main sections and choruses, and a final section. Instead of generating audio directly, the system represents music symbolically and applies selection, crossover, and mutation to improve compositions according to a heuristic fitness function based on musical criteria. The objective is to assess whether this process can generate diverse, musically coherent pieces while preserving the structural constraints of the proposed representation.

2. RELATED WORK

Diverse recent studies have explored evolutionary approaches to symbolic music generation, investigating different representations and strategies.

Poćwiardowski, et al. [1] propose $M^6(GPT)^3$, a system for generating multitrack, multi-minute MIDI music by combining genetic algorithms, probabilistic methods, and GPT models. It uses a structured symbolic representation to organize musical components and support their evolutionary generation. The work demonstrates how evolutionary algorithms can produce longer, editable MIDI compositions. It is directly related to the present study through its use of MIDI and genetic algorithms, although it adopts a broader approach that also incorporates language models and probabilistic methods.

Soong et al. [2] present a genetic algorithm for monophonic music generation using a non-binary genetic representation to encode musical elements. Their work explores how gene structures represent musical characteristics and enable evolutionary operators to generate compositions. This music-oriented representation is relevant because it supports constraints and guides the search toward musically meaningful solutions. Unlike their monophonic approach, the present work aims to generate MIDI compositions combining musical elements.

Lopez-Rincon et al. [3] propose a music generation method based on genetic recombination of harmonic structures. Using symbolic music representation, their approach combines existing musical elements and applies harmonic criteria to guide evolution. This work demonstrates how higher-level musical information can guide generation beyond random note selection. Similarly, the present study incorporates music-theory constraints into the generation and evaluation of MIDI individuals.

The reviewed works demonstrate different applications of genetic algorithms to symbolic music composition, including encoding of musical elements, recombination of harmonic structures, and generation of multitrack MIDI music. These studies motivate the use of genetic algorithms in the present work, which similarly adopts an evolutionary approach to generate symbolic music.

3. METHODOLOGY

A genetic algorithm was chosen due to its recent applications to music generation and its ability to iteratively improve candidate compositions through evolution. Its population-based search provides a suitable framework for generating MIDI under predefined musical constraints.

3.1 Genotype

The genotype was designed as a structured representation of the composition, using independent fixed patterns for each musical component rather than encoding the entire song as a single sequence of notes. Drum patterns represent each bar as a sequence of time-step genes, with each gene containing a variable set of percussion elements. Harmonic information is represented by main chords, each

defined by a root, chord type, and inversion. Bass patterns encode sequences of notes within a bar, where each note is defined by its position and duration, while arpeggio patterns define the number of notes, their rhythmic step, and their ordering within the bar. Finally, melodic patterns span eight bars and represent notes through their pitch, velocity, position, and duration. These constrained representations ensure that randomly initialized genotypes produce valid musical patterns while maintaining sufficient variability for the evolutionary process.

3.2 Crossover

The crossover operator recombines two parent individuals while preserving the structural validity of the genotype. Drum sequences use a multi-point crossover with independently selected cut points for each pattern. Bass patterns are recombined by selecting or regenerating valid one-bar patterns, while arpeggio patterns combine their note count, rhythmic step, and note ordering. Melodic patterns are recombined using a temporal cut that combines notes from both parents. For harmonic information, each chord is selected from the parents' chord sets, with a probability of introducing a new randomly generated chord to increase diversity. All resulting components are independently copied to form the child individuals.

3.3 Mutation

The mutation operator independently modifies the different components of each individual to maintain genetic diversity and reduce premature convergence. Drum genes can be regenerated, removed, extended, or have their percussion elements replaced, while harmonic elements can undergo changes in root, chord type, and inversion. Bass and arpeggio patterns are mostly replaced by newly generated valid patterns, with occasional smaller modifications to arpeggio ordering. Melodic patterns can undergo note insertion, removal, or changes in pitch, velocity, position, and duration. Each component is controlled by its own mutation probability, allowing different parts of the musical structure to evolve at different rates.

3.4 Fitness

The fitness function combines multiple subfitness functions, each evaluating a musical characteristic of the composition. The rhythmic component assesses strong beats, backbeat, kick and hi-hat patterns, syncopation, density, variation, internal structure, and instrument combinations, while additional scores evaluate the chorus and introductory and ending drum patterns. The harmonic component assesses chord validity, tonal coherence, chord quality, root motion, voice leading, main-section stability, chorus characteristics, and harmonic diversity. Bass and arpeggio patterns are evaluated by their rhythmic structure, with additional diversity scores for arpeggios. Melodic patterns are assessed by note density, pitch intervals and range, rhythm, duration, dynamics, temporal coverage, and diversity between melodies. Finally, melodic-rhythmic alignment measures the relationship between melody and drum patterns. These scores are combined using predefined weights at different hierarchical levels to produce a final fitness value normalized to $[0, 1]$, guiding evolution toward musically coherent and diverse compositions.

3.5 Population Evolution

The initial population is composed of a predefined number of individuals, each initialized using the proposed genotype representation and probabilistic musical patterns. At each generation, all individuals are evaluated using the fitness function, and new individuals are selected through tournament selection, in which three randomly chosen candidates compete and the one with the highest fitness is selected. Elitism preserves the best individual from the previous generation, while crossover and mutation are applied to generate the remaining individuals, with a small proportion of random immigrants introduced to maintain population diversity. The algorithm uses a fixed number of generations as its stopping criterion. During evolution, the minimum, mean, generation best, and global best fitness values are recorded and reported in the fitness evolution graph, allowing the progression and convergence of the population to be analyzed.

3.6 Final Midi Composition

The MIDI generation organizes the composition into six fixed sections: *Intro*, *Base1*, *Chorus1*, *Base2*, *Chorus2*, and *Ending*, as exemplified in Figure 2 following a traditional popular Brazilian music. The drum track is the first to be added to the composition and is mandatory, providing the rhythmic foundation for the entire piece. Different drum patterns are used for the introduction, main sections, choruses, and ending, allowing the rhythmic character to vary according to the musical section.

The harmonic structure is defined by the main chords, which are mandatory elements of the composition. The chords are assigned to the string tracks and provide the harmonic reference for the other instruments. String1 plays the complete chord voicings, while String2 is introduced later and reproduces the harmonic material in a higher intensity. Bass1 follows the active main chords using the corresponding bass patterns, and Bass2 reinforces the lowest chord tones one octave below String1.

The remaining instrumental tracks are introduced progressively to increase the variation and density of the arrangement. Guitar1 performs arpeggios derived from the main chords and enters after an initial period of silence, while Guitar2 is introduced later by reproducing the arpeggiated material one octave higher. This gradual introduction of instruments avoids having all tracks active from the beginning and creates changes in the texture throughout the musical structure.

The melody is generated from the two melodic patterns and can be optionally aligned with drum onsets and adjusted to the active main chords. In contrast to the other tracks, greater variation is deliberately allowed in the melody, including deviations from the exact rhythmic

or harmonic patterns, in order to produce a less rigid and more realistic musical result. From the second half of the composition, a second melodic voice is added using lower chord tones, complementing the main melody and increasing the musical density of the later sections.

3.7 Wave and Video Rendering

At the end of the process, the generated composition is saved as a MIDI file, providing the symbolic representation of the final result. A piano-roll image is also generated to visually represent the notes and their temporal organization. The MIDI file is converted into a WAV audio file using the *PC51d.sf2* SoundFont, allowing the generated composition to be auditioned. Finally, an MP4 video is created by combining the resulting audio with the piano-roll image as the background, providing an audiovisual representation of the generated piece.

4. EXPERIMENTS

For the reported experiments, the musical configuration was kept fixed, with the Acoustic Grand Piano assigned to the melody track and all other instruments unchanged, following their track names.

4.1 Genetic Algorithm Tuning

Table 1 presents the tested configurations, including population size (Pop.), number of generations (Gen.), crossover rate (Cross.), mutation rate (Mut.), and elitism (Elit.), along with the best obtained fitness values (BFit.). The results indicate that elitism improves the best fitness achieved, while a higher crossover rate combined with a lower mutation rate also yields better results. Increasing the population size and number of generations further improves the best fitness, as expected for a classic genetic algorithm. Overall, the experiments highlight the influence of population size and genetic operators on the quality of the generated compositions.

Table 1. Genetic algorithm main experiments.

Exp.	Pop.	Gen.	Cross.	Mut.	Elit.	BFit.
1	10	10K	0.5	0.5	No	0.849
2	10	10K	0.8	0.1	No	0.880
3	10	10K	0.5	0.5	Yes	0.867
4	10	10K	0.8	0.1	Yes	0.889
5	20	20K	0.8	0.1	Yes	0.898

Figure 1 shows the evolution of fitness throughout the genetic algorithm, using the best parameter configurations presented in Table 1. The best, mean, and worst fitness values exhibit an overall improvement over generations, indicating progress in the quality of the population. Although the mean and worst fitness values vary between generations, their overall behavior, together with the steady increase in the global best fitness, suggests that the evolutionary process progressively improves the candidate compositions while maintaining diversity within the population.

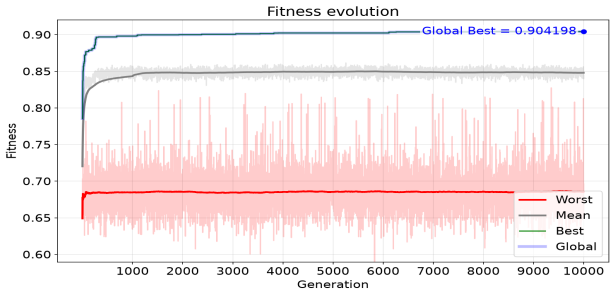


Figure 1. Example of evolution of the Genetic Algorithm.

4.2 Example of Generated Music

Figure 2 shows a generated composition in piano-roll format. The *Drums* track identifies the six sections: *Intro*, *Base1*, *Chorus1*, *Base2*, *Chorus2*, and *Ending*. The *String1* track contains four chords for the main sections and four for the choruses. *String2* enters midway, reproducing the first track in a higher register. *Bass1* and *Bass2* reinforce the harmony, while *Guitar1* and *Guitar2* progressively introduce arpeggios based on the active chords. Finally, the *Melody* track represents the main vocal melody, with a second voice entering in the second half. This overall musical structure remains fixed across all generated compositions. However, the genetic algorithm evolves the internal musical content, exploring variations in chord progressions, drum patterns, and the timing and distribution of notes across the different tracks.

The source code is publicly available on GitHub [1], where several example compositions are provided, including the corresponding genetic algorithm fitness evolution plots generated during their creation.

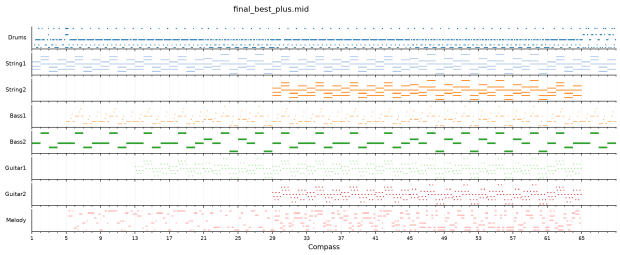


Figure 2. Example of a generated midi composition.

5. CONCLUSIONS

The proposed genetic algorithm generated complete compositions inspired by Brazilian popular music, with considerable variation among the pieces. The structured genotype, musical constraints, and fitness components contributed to harmonic, rhythmic, and melodic coherence. However, the fixed song structure and music-theory rules limit flexibility and stylistic diversity. Future work could expand the genotype with additional chords, instruments, and rhythmic patterns, while incorporating global musical structure to support more song forms.

[1] `REPOSITORY_LINK`

6. REFERENCES

[1] J. Poćwiardowski, M. Modrzejewski, and M. S. Tatara, "M6(GPT)3: Generating multitrack modifiable multi-minute MIDI music from text using genetic algorithms, probabilistic methods and GPT models in any progression and time signature," in *2025 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*, 2025, pp. 1–6.

[2] H.-C. Soong, R. K. Ayyasamy, C. S. Ku, and S. Khamis, "Evolutionary music monophony generator using non-binary representation genes from genetic algorithm," in *Machine Learning and Intelligent Systems*, vol. 393. IOS Press, 2024, pp. 31–40.

[3] O. Lopez-Rincon, O. Starostenko, and A. Lopez-Rincon, "Algorithmic music generation by harmony recombination with genetic algorithm," *Journal of Intelligent and Fuzzy Systems*, vol. 42, no. 5, pp. 4411–4423, 2022.